

ODIN: Optomechanical distributed instrument for inertial sensing and navigation

Jose Sanjuan

Wyant College of Optical Sciences
University of Arizona
Tucson, USA
jsanjuan@arizona.edu

Jackson Dahn

Wyant College of Optical Sciences
University of Arizona
Tucson, USA
jdahn@arizona.edu

Ramses Miranda-Espino

Wyant College of Optical Sciences
University of Arizona
Tucson, USA
mirandaeramses@arizona.edu

Saruul Nasanjargal

Wyant College of Optical Sciences
University of Arizona
Tucson, USA
saruul@arizona.edu

Andrea Nelson

Wyant College of Optical Sciences
University of Arizona
Tucson, USA
ORCID: 0000-0002-5909-4692

Jeremiah Stoddart

Wyant College of Optical Sciences
University of Arizona
Tucson, USA
bostod@arizona.edu

Mohanad Warrayat

Wyant College of Optical Sciences
University of Arizona
Tucson, USA
mohanadwarrayat@arizona.edu

Miguel Dovale

Wyant College of Optical Sciences
University of Arizona
Tucson, USA
mdovale@arizona.edu

Felipe Guzman

Wyant College of Optical Sciences
University of Arizona
Tucson, USA
felipeguzman@arizona.edu

Abstract—ODIN (Optomechanical Distributed Instrument for Inertial Sensing and Navigation) is a novel low-frequency optomechanical inertial sensing system funded under NASA’s INVEST program for space technology demonstration. The instrument employs twelve uniaxial fused silica optomechanical accelerometers arranged in a gyroscope-free cubic configuration, enabling simultaneous recovery of linear acceleration, angular acceleration, and centripetal terms. ODIN targets sensitivities of $\sim 10^{-10} \text{ m s}^{-2}/\sqrt{\text{Hz}}$ and $7 \times 10^{-10} \text{ rad s}^{-2}/\sqrt{\text{Hz}}$ at 10 mHz for linear and angular acceleration, respectively. ODIN will fly as a secondary payload aboard the GRATTIS (Gravitational Reference Advanced Technology Test in Space) mission, scheduled for launch in early 2027, representing a significant step toward compact, cost-effective space-based accelerometry for future Earth geodesy and inertial navigation applications.

Index Terms—Geodesy, inertial navigation, accelerometers, optomechanics, space, interferometry

I. INTRODUCTION

Accelerometry and inertial navigation are an essential aspect of applications related to space geodesy [1] and applications where information about localization and orientation is necessary [2]. The former is especially critical in space geodesy, where precise measurements of non-inertial forces due to e.g. residual drag, radiation pressure, and spacecraft thrusters are necessary in the data analysis for gravity field determination. The latter has seen rapid growth in recent years, driven in particular by the expanding field of autonomous and unmanned vehicles operating in environments where global navigation satellite systems (GNSS) or pre-existing maps are unavailable. Inertial navigation systems (INS) traditionally rely on triaxial

accelerometers paired with gyroscopes to measure the linear acceleration and angular velocity of a rigid body, respectively. An alternative approach uses an array of linear accelerometers alone to recover both linear and angular acceleration, a class of systems known as gyroscope-free inertial navigation systems (GF-INS) [3], [4]. Compared to conventional INS architectures, GF-INS designs offer reduced complexity, cost, mass, volume, and power consumption, while also improving reliability. The use of a single sensor type further simplifies system integration, interfacing, and calibration.

Here we present ODIN, the Optomechanical Distributed Instrument for Inertial sensing and Navigation for space applications, based on low-frequency optomechanical accelerometers in a gyroscope-free configuration. ODIN is part of the GRATTIS mission [5], a NASA-funded technological demonstrator with a launch scheduled for early 2027. ODIN will share an Apex Aries bus with the primary payloads, two simplified gravitational reference sensors (S-GRS), an electrostatic accelerometer for future gravity field missions, developed by the University of Florida [6]. The spacecraft will be in a polar orbit at an altitude of $\sim 550 \text{ km}$. ODIN aims to demonstrate sensitivities at 10 mHz of $\sim 10^{-10} \text{ m s}^{-2}/\sqrt{\text{Hz}}$ for linear acceleration, and $7 \times 10^{-10} \text{ rad s}^{-2}/\sqrt{\text{Hz}}$ for angular acceleration, performance levels that would establish ODIN as a compact yet highly sensitive instrument suitable for deployment as a primary or redundant accelerometer and/or inertial sensor in future space geodesy missions. To this end, the primary mission objective is to characterize the instrument’s noise floor and validate its ability to recover spacecraft body

accelerations in an orbital environment representative of future geodesy mission requirements.

ODIN is comprised of six inertial sensing frames each consisting of two monolithic fused silica resonators with axes oriented 90° apart and a fundamental mode of ~ 8 Hz for a total of twelve sensing axes. These six frames make up the six sides of a cubic configuration, allowing for full sensing capability. This system is an expansion of our uniaxial and dual-axis linear optomechanical accelerometers, which are comprised of a linearly oscillating fused silica test mass on the end of two leaf spring flexures with a heterodyne interferometric readout.

II. OPTOMECHANICAL ACCELEROMETERS

The core sensing elements of ODIN are optomechanical accelerometers based on fused silica mechanical resonators [7]. In each resonator, a test mass is suspended by two parallel leaf flexures approximately $100\ \mu\text{m}$ thick and a few centimeters in length. When operated in a strapped-down configuration, non-gravitational accelerations, δa , acting on the spacecraft induce displacements, δx , on the test mass, which are measured and converted to acceleration via the resonator's transfer function. Below its resonance frequency, f_0 , it is simply

$$\delta a = \frac{1}{(2\pi f_0)^2} \delta x, \quad (1)$$

where δa represents the non-gravitational accelerations on the accelerometer, i.e., on the spacecraft, δx the displacement of the test mass due to the acceleration, and f_0 is the resonance frequency of the resonator.

Figure 1 shows a dual (two resonators on a single wafer) optomechanical accelerometer prototype. The resonators, fabricated from a single wafer via laser-assisted etching, exhibit large quality factors Q (on the order of 10^5), which together with test masses weighing several grams yield mQ -product values above 1,200 kg (a key metric for achieving high acceleration sensitivities). The fundamental noise limit of such sensors is defined by the thermal noise, which can be estimated as

$$S_{a_{\text{th}}}(f) = \frac{4k_B T 2\pi f_0^2}{mQf}, \quad (2)$$

where k_B is Boltzmann's constant, T is temperature in Kelvin, m is mass of the oscillator, Q is the mechanical quality factor, and f_0 is the resonant frequency. $S_{a_{\text{th}}}(f)$ is expressed in terms of the power spectral density, i.e., in units of $(\text{m s}^{-2})^2/\text{Hz}$. Predicted thermal noise limit is shown in Fig. 3 (oher trace).

A key aspect of accelerometer performance is how well the test mass motion can be measured. In this regard, low resonance frequency of the resonator alleviates the requirements on the displacement sensing. Still, for resonant frequencies in the order of few Hz, sub-picometer precision is required to reach the thermal noise limit. Displacement sensing is achieved through a dual heterodyne interferometer architecture that provides common-mode noise rejection [8]. One interferometer tracks the motion of the test mass while a second, reference interferometer measures a fixed mirror. Taking the differential signal preserves the test mass motion

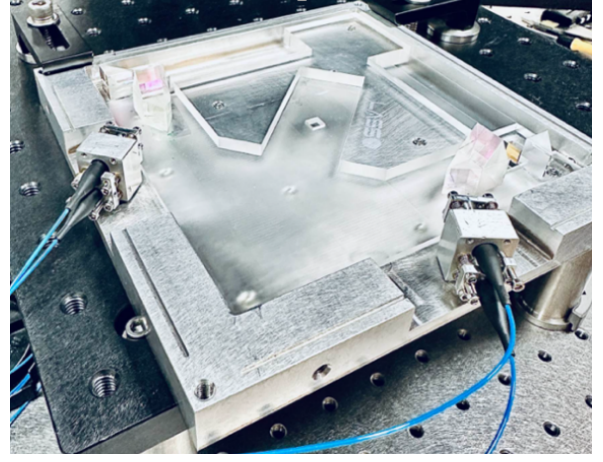


Fig. 1. Dual optomechanical resonator prototype.

while suppressing phase and frequency noise common to both paths, including contributions from the laser and fiber links. The system uses two frequency-shifted input beams derived from a pair of 1550 nm diode lasers, producing a heterodyne beat frequency in the MHz range.

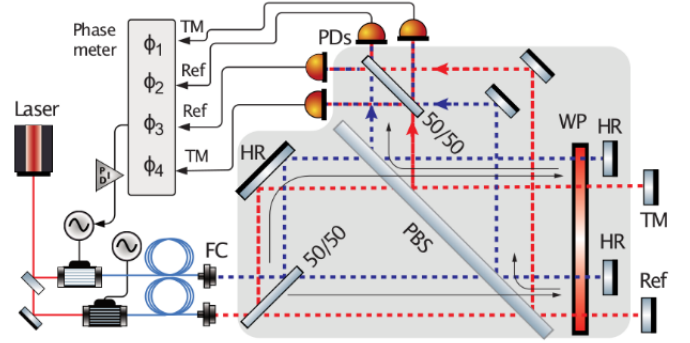


Fig. 2. Dual heterodyne interferometer schematic including balanced detection [1]. Two laser beams with frequencies MHz together with polarizing optics create two interferometers: the reference interferometer, and the test mass interferometer. The differential signal cancels out common phase noise while retaining the signal of interest: the test mass motion. TM: test mass, Ref: reference, PBS: polarizing beam splitter, HR: high reflectivity; WP: (quarter) waveplate.

Figure 3 shows the performance of the optomechanical accelerometers in the laboratory environment. The red trace corresponds to a measurement where the resonators are freely moving and thus dominated by seismic activity. The blue trace shows the measurement provided by a nearby commercial seismometer (T120 Trillium), which shows an excellent agreement with the optomechanical accelerometer. The green trace shows the noise attributed to the displacement sensing noise, i.e., the noise of the interferometer converted to acceleration since here the resonator is anchored. Finally, the ultimate theoretical thermal noise limit of the resonator is shown by the oher trace.

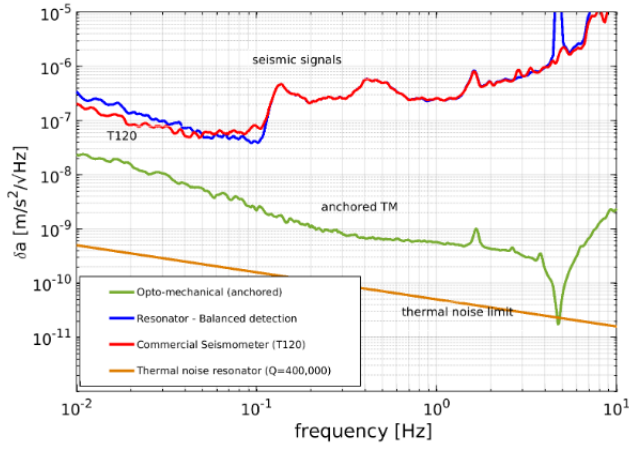


Fig. 3. Optomechanical resonator performance expressed in the frequency domain in terms of the square root of the power spectral density.

III. ODIN AND THE GYRO-FREE INERTIAL NAVIGATION CONFIGURATION

ODIN's gyro free configuration is shown in Fig. 4 where the colored dots represent accelerometers' positions, and the arrows sensing directions. Each side of the cube configuration is made by the dual axis accelerometers described previously. Note each accelerometer only senses one direction since they are uniaxial. The signals measured by each accelerometer are:

$$A_i = [\mathbf{a}_{sc} + (\dot{\boldsymbol{\Omega}}_{sc} + \boldsymbol{\Omega}_{sc}^2) \cdot \mathbf{r}_i + n_i], \quad i = 1 \dots 12 \quad (3)$$

where $\mathbf{a}_{sc}$, $\dot{\boldsymbol{\Omega}}_{sc}$, $\boldsymbol{\Omega}_{sc}$, $\mathbf{r}_i$, and n_i are the spacecraft linear acceleration, angular acceleration, angular velocity, position vector of the i -th accelerometer with respect to the spacecraft center of mass, and accelerometers' noise, respectively. The $\cdot$ indicates the dot product between acceleration and sensing direction.

For the configuration shown in Fig. 4, it can be shown that linear and angular accelerations, in addition to the six centripetal terms, can be estimated as [4]

$$\hat{\mathbf{s}}^T = \mathbf{J}^{-1} \mathbf{A}^T \quad (4)$$

where $\hat{\mathbf{s}} = [\hat{a}_x, \dots, \hat{\Omega}_x, \dots, \hat{\Omega}_x^2, \dots]$ are the state estimates ($\hat{\cdot}$ here indicates estimates), $\mathbf{A} = [A_1 \dots A_{12}]$ are the accelerometers' measurements, and $\mathbf{J}$ is a 12×12 matrix purely defined by the geometry of the configuration, i.e., by the positions and sensing directions of the accelerometers.

One of the primary goals of ODIN on the GRATTIS mission is to demonstrate instrument performance in orbit. To do so, one must be able to measure the inherent noise of the accelerometers. However, environmental disturbances such as residual drag, radiation pressure, and Euler and centripetal accelerations are expected to be significantly larger than the intrinsic noise floor of the optomechanical accelerometers. Fortunately, a linear combination of accelerometer outputs exists that cancels all of the the above signals while preserving the inherent accelerometers noise:

$$\mathbf{O} = \mathbf{x}^T (\mathbf{A} - \mathbf{J} \hat{\mathbf{y}}_*^T) \quad (5)$$

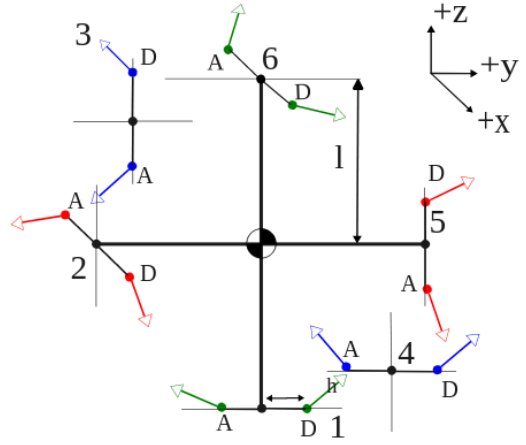


Fig. 4. ODIN's gyro-free configuration. A total of twelve uniaxial accelerometers are used to retrieve 12 variables (linear and angular acceleration, and centripetal acceleration).

where $\mathbf{x}$ is an arbitrary vector and $\hat{\mathbf{y}}_*^T$ represents the estimated accelerations from Eq. (4) but substituting the centripetal terms with those derived from angular accelerations through time integration. A key aspect for Eq. (5) to be useful to determine accelerometers' noise levels is the knowledge of positioning and orientation of the accelerometers. For this reason, precise alignment and positioning of the six sides of the cube with coordinate measuring machine (CMM) metrology is necessary, together with stringent requirements in spacecraft attitude control since the latter couples with position/orientation uncertainty to generate noise. Figure 5 shows ODIN's engineering development unit (EDU) fused silica wafer with four metrology spheres and frame in place.



Fig. 5. ODIN's EDU dual resonator. Each of the six wafers includes metrology spheres for accurate relative alignment between the accelerometers.

A key design upgrade relative to the laboratory prototypes is the incorporation of launch locks to secure the test masses and flexures during launch. These are implemented by axially pre-loading the flexures using a piezo motor, which not only immobilizes the test mass but also shifts the flexure violin

modes to higher frequencies where spacecraft and launcher vibrations are significantly reduced. Figure 6 shows a CAD model where the launch locks (in the bottom side) can be seen. The model also shows the interferometer and the photodiodes on both sides for a total of four photodiodes (two per accelerometer) per wafer.

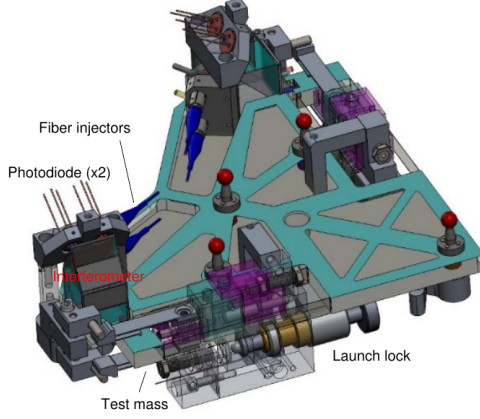


Fig. 6. ODIN's sensor CAD model. Six of these units are arranged in the cube-like configuration shown in Fig. 4. The model shows the fiber injectors (two per side), the interferometers (one per side), the launch locks (one per side), and the photodiodes (two per side).

ODIN's instrument architecture comprises two distinct packages, as shown in Fig. 7. The sensor package, assembled and integrated by Ruda Optical, houses the resonators, free-beam optics, photodiodes, and launch locks. The opto-electronics package, developed by IST Precision, contains the transimpedance amplifiers (TIA), fiber optics, two lasers, a microcontroller, and a field-programmable gate array (FPGA). The FPGA implements a 24-channel IQ demodulation phasemeter and a frequency-offset phase-locked loop that maintains the two laser frequencies ~ 1 MHz apart, as required by the heterodyne readout architecture.

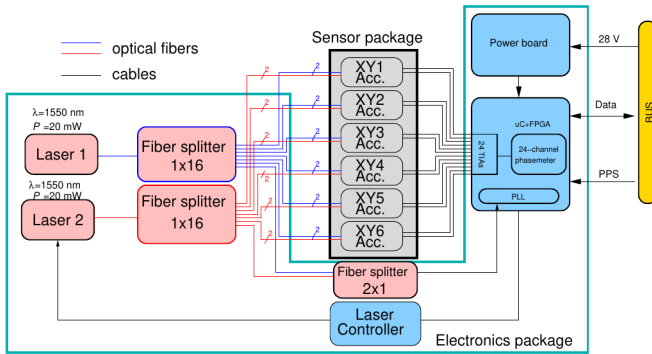


Fig. 7. ODIN's high-level architecture.

IV. SUMMARY AND CURRENT STATUS

ODIN's engineering development unit (EDU) dual-axis resonators are currently under characterization, including resonance frequency, quality factor, alignment, and noise floor

measurements. Environmental testing, covering survivability under maximum predicted launch loads and the expected thermal environment, is planned for the coming weeks. Predicted in-orbit performance is shown in Fig. 8, with the left panel showing linear acceleration sensitivity and the right panel showing angular precision.

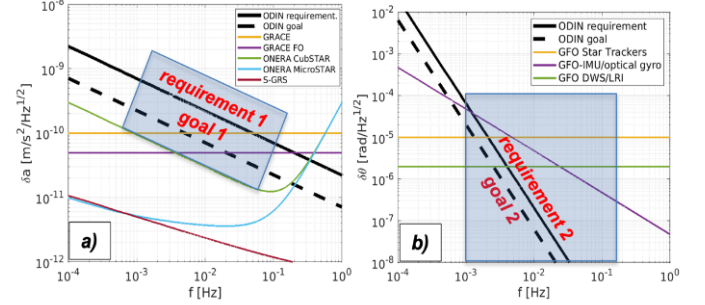


Fig. 8. ODIN's predicted in-orbit performance.

Successful on-orbit demonstration of ODIN will validate a new class of compact, high-sensitivity inertial sensors with broad applicability. The gyroscope-free optomechanical architecture enables cost-effective mission designs, spacecraft miniaturization, and simplified system integration. The large dynamic range of optomechanical accelerometers further supports deployment in challenging environments, including satellite constellations operating at lower altitudes or in the presence of large non-gravitational disturbances such as thruster firings.

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